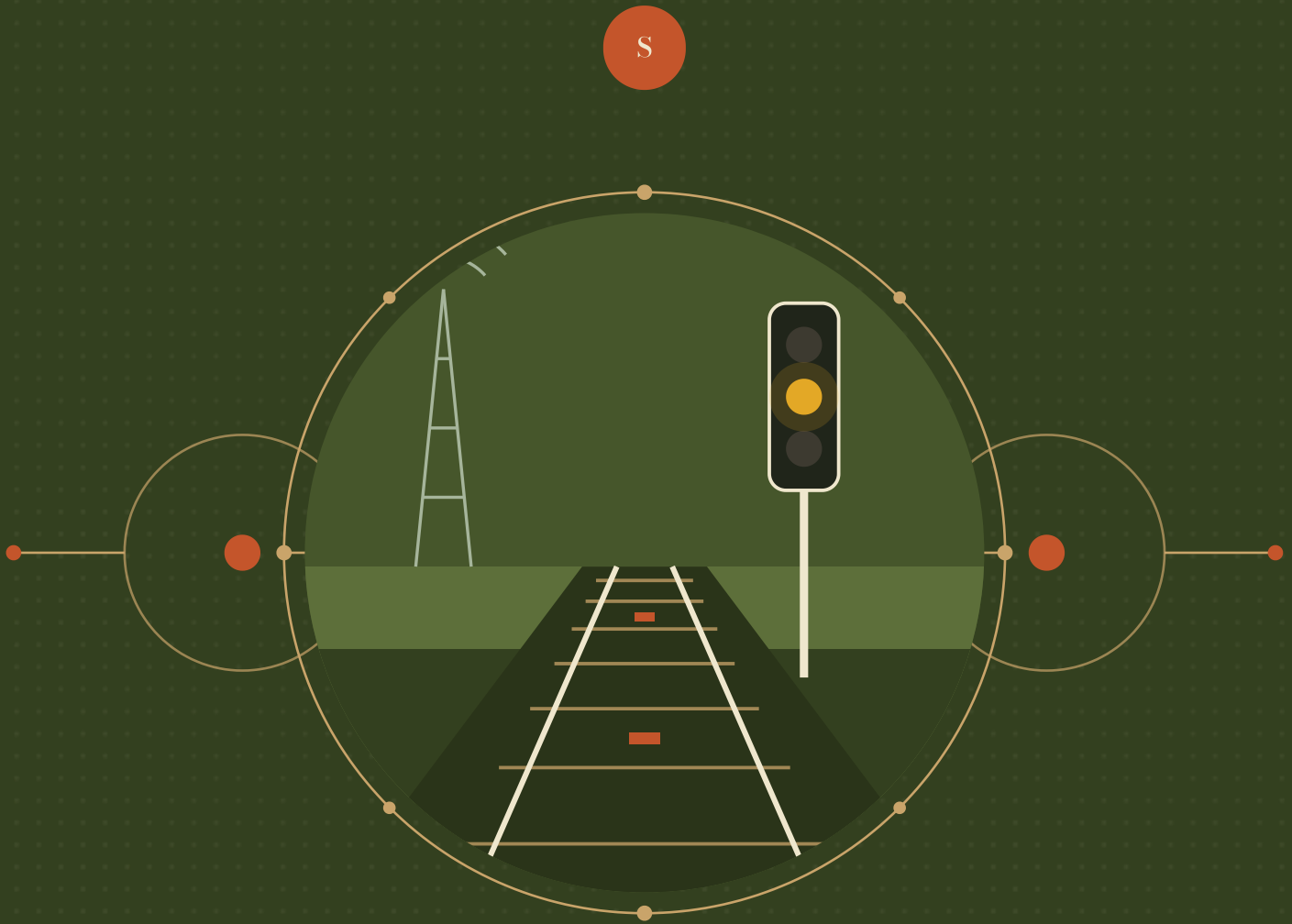




SignalSafe

A browser-based digital twin of the Kavach Generic Test Bench: build a yard, run Onboard and Stationary Kavach through it, inject the faults the field throws at them, and get a clause-by-clause verdict on what the system did and why.

DOCUMENTATION FIRST · PRE-BUILD

GROUNDING IN

Kavach SRS v4.0 Amdt-3

RDSO/SPN/196/2020 · eff. 07.06.2024

MODELLED ON

Draft FRS, Generic Test Bench

RDSO/GTB Kavach/FRS/2026 · Jul 2026

STATUS

Design draft

Build starts after sign-off

What's *inside*

01	Project overview	the problem, the idea, who it's for
02	Kavach on one page	the system SignalSafe simulates
03	What we'll build	every module, mapped to the RDSO test bench
04	The scenario library	twelve clause-grounded test scenarios
05	Faults & evaluation	what we can break, and how we judge it
06	Screens we'll design	wireframes of the four main views
07	Tech stack	frontend, engine, data, reports
08	Architecture	layers, timing, data model
09	Engineering principles	determinism, fail-safety, honesty
10	Build plan	eight weeks, milestone by milestone
11	Future features	what could come after v1
A	Appendix: glossary, traceability, design language	

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BENCH MODULES

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SEED SCENARIOS

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INJECTABLE FAULTS

2 s

RADIO FRAME CYCLE

How to read this document

This is written **before any code**, so it is a specification and plan, not a report. Clause numbers in rust (for example **SRS 20.1.2** or **FRS 7.6.7**) point to the two RDSO documents listed in the appendix; every number SignalSafe enforces is traced to one of them. Items tagged **MVP** are committed for v1. **V2** and **LATER** items are planned but not promised.

A Kavach test lab that *fits in a browser tab*

SignalSafe is a software twin of the reference test facility RDSO describes for Kavach. A user lays out a station yard, places signals, points, RFID tags and radio towers, runs simulated locos through it, and injects faults: radio loss, a missed tag, a GPS failure, a signal flickering on the approach. SignalSafe then checks each response against the SRS and explains the chain of events.

The problem

- Kavach function testing needs a physical bench: real SVK and OVK racks, relay adapters for field inputs, radio and brake simulators. Few people outside OEMs and RDSO ever get to use one.
- RDSO's July 2026 draft FRS asks for a generic test bench but notes that OEMs don't share common simulator or adapter interfaces, and that *"further study is required"* **FRS 7.6.1.18**.
- The SRS is dense: timeouts, fallback modes and brake rules are spread across 34 sections. Students and new engineers find it hard to see how they combine in a real incident.

Our answer

- A **browser-based bench** with the same module breakdown as the FRS: Scenario Editor, Bench Controller, Yard, Loco, Radio, DMI and SMOCIP simulators, Scenario Logger and Evaluation Tool.
- **Reference models** of Onboard and Stationary Kavach, written from the SRS, that stand in for the equipment under test.
- A **generic adapter contract**, so the same scenarios could one day drive real equipment over serial or Ethernet.
- An **evaluation report** that cites the clause behind each pass or fail.

Who it's for

Students and educators in railway signalling and safety-critical systems, who need to see Kavach logic run rather than only read it. **Test engineers** drafting scenario sequences, which the FRS places in the bench's scope **FRS 6.1.1.6**. They can prototype and review these sequences before booking time on a hardware bench. **Reviewers** comparing OEM behaviour against a neutral reading of the SRS.

A test engineer's journey

1 · BUILD YARD	2 · WRITE SCENARIO	3 · PRE-SET	4 · RUN	5 · EVALUATE	6 · REPORT
Tracks, points, signals, TINs, RFID tags, LC gates, towers.	Trains, configs, route timeline, faults, expected observables.	Load config into reference SVK/OVK, power on, register.	Watch live: yard, DMI, SMOCIP, radio log. Step, pause, speed up.	Observables matched in time and location windows.	Pass/fail per clause, causal chain, exportable PDF.

Steps 3–5 follow the simulation phases in **FRS 8.2.2**: preparation, start and sync, initial communication, scenario run, and simulation stop with event-logger download.

The diagram illustrates the SVK handover process between Station A (SVK-A) and Station B (SVK-B) for a train in the Block Section (TIN 201). The train is shown with a green locomotive (LOCO 37421) and orange passenger cars (OVK). The process involves UHF communication every 2 seconds between the SVK units and the train. The handover occurs at the end of the block section, marked by a red dot labeled S1. The diagram also shows the train's position relative to the block section boundaries and the handover point.

STATION A · SVK-A

BLOCK SECTION · TIN 201

STATION B · SVK-B

SVK-A · 2002

SVK-B · 2002

SVK ↔ SVK handover (Annex P)

UHF · every 2 s

S4

S1

RFID tag pairs · ≤1000 m apart

LOCO 37421 · OVK

MOVEMENT AUTHORITY → EOA AT S1

OVK reads tags → reports location every 2 s → SVK reads interlocking → returns aspect, MA, track profile → OVK builds brake curve, brakes if exceeded.

Trackside

- **RFID tags**, duplicated, on the sleepers. They carry absolute location and TIN, and correct odometry drift (5 m + 5% of distance since the last tag) **SRS 3.4.2.4**.
- **Stationary Kavach (SVK)**, a 2-out-of-2 vital computer. It reads interlocking inputs (signals, points, track circuits) and generates movement authority **SRS 3.4.4**.
- **SMOCIP**, the station master's panel with SoS button, health indication and fault codes **SRS 3.4.6**.
- **Remote Interface Units** bring remote signals in over OFC **SRS 3.4.7**.

Onboard

- **Onboard Kavach (OVK)**, a 2-out-of-2 vital computer that supervises speed against MA and the brake curve **SRS 3.5.2.**
- **LP-OCIP (DMI)**, the driver display with speeds, target distance, mode and soft keys. SoS is sent by pressing SoS + Ack together **SRS 3.5.5.**
- **BIU**, which issues Normal, Full-Service and Emergency brake commands and cuts traction **SRS 3.5.6.**
- **Two RFID readers, two radios, two pulse generators**, all in hot standby.

Operating modes SignalSafe models

Mode names as used in the SRS and FRS. The full transition table is in Annexure A1, which is not in the documents we have. Section 09 explains how we handle that gap.

Full Supervision (FS)

Full MA, signal aspect and track profile known. OVK supervises against the full brake curve.

On Sight (OS) • Limited Supervision (LS)

Reduced authority. The loco pilot acknowledges the transition with Common/Ack **SRS 3.5.5.7**.

Staff Responsible (SR)

The degraded fallback, e.g. after GPS time-out **SRS 20.3.5** or loss of the TSRMS, adjacent-SVK or EI link **SRS 20.8**.

Trip · Post Trip · Override

Entered after passing an EOA. Post-Trip and Override are selected on the DMI **FRS 7.6.1.12**.

Standby • Isolation • Non-Leading

Not supervising, BIU isolated, or trailing loco. No LC auto-whistle in these modes **SRS 15.7**.

System Failure (SF)

Safe state after a vital failure: both RFID readers, PG mismatch, active DML, brake feedback, or GPS + RTC **SRS 20.2-20.6.**

Every module of the *RDSO bench*, in software

The draft FRS splits the Generic Test Bench into the equipment under test, a test adaptor layer, and fourteen bench modules (Figures 1–3, section 7.6). SignalSafe keeps that split exactly, so anyone who knows the FRS can find their way around. The table shows each module and where we stand.

FRS MODULE	WHAT IT DOES IN SIGNALSAFE	FRS REF	PLAN
Test Scenario Editor (TSE)	Create, edit, delete and save scenarios. A database of test cases that combine into sequences. Correctness checks flag a signal with no foot tag, a tag gap over 1000 m, or a route not in the control table.	7.6.2	MVP
Test Bench Controller (TBC)	Pick a scenario, start or stop it, power SVK/OVK up or down, inject System Failure. Live panel shows yard state, speed, location, last tag, DMI, SMOCIP and radio traffic.	7.6.3	MVP
Yard Simulator	Simulated interlocking: signal ECRs, point WKR, track circuits, LC gates, line-clear. Drives SVK field inputs and can flicker or fail any input.	7.6.13	MVP
Speed Simulator (SS)	Train dynamics: traction, gradient, brake response to BIU commands. Feeds odometry and holds within 1% of theoretical speed 7.6.5.5.	7.6.5	MVP
RFID Simulator (RFID-S)	Fires tag reads from absolute location, with no accidental misses up to 200 km/h. Can drop one or both tags of a pair, or plant an unexpected tag.	7.6.6	MVP
Radio Message Simulator (RMS)	Carries OVK↔SVK packets on the 2 s frame. Adds delay, packet loss, radio holes and connection loss at set locations. Emulates two radios per unit.	7.6.7	MVP
BIU Simulator (BIU-S)	Receives NSB, FSB, EB and traction cut-off. Returns brake-pipe and brake-cylinder pressure. Can fail pressure feedback.	7.6.4	MVP
DMI Simulator (DIS)	An on-screen LP-OCIP: speeds, target distance, MA, aspect, mode, soft keys, SoS / Ack / Cancel. Can play the scenario's inputs automatically, with no operator.	7.6.8	MVP
SMOCIP Simulator	Station master panel: SoS button, health LED, 7-segment fault codes.	7.6.12	MVP
Test Scenario Logger (TSL)	The bench's single source of truth. Every module message is logged with sim time and location, alongside both units' event loggers.	7.6.11	MVP
Test Evaluation Tool (TET)	Offline matcher: finds each expected observable in its time and location window, avoids double matches, and builds the causal chain.	7.6.15	MVP
GPS Simulator (GPSS)	Time source for both units. Fails GPS1, GPS2, PPS or RTC independently, following the SRS GPS truth table.	7.6.9	V2
Power Analyzer (PA)	Switches any module of SVK or OVK on or off and injects supply faults.	7.6.10	V2
Track Profile Simulator (TPS)	Edits PSR, TSR, gradient and LC data inside the SVK's application logic during a run. The FRS notes its requirements are still to be explored 7.6.14.3.	7.6.14	V2

Beyond the FRS

- **Adjacent SVK simulator:** up to six neighbours for handover tests **FRS 7.4.9.3**. **V2**
- **Causal "why" view:** every log event records the event that caused it, so a failed observable can be traced back to the injected fault. **MVP**
- **Replay:** any run can be replayed exactly from its seed and inputs. **MVP**

Explicitly out of scope

- Card-level hardware testing, which is also outside the FRS's scope **FRS 5.1.1.5**.
- Key-management cryptography. The SIM-based KMS isn't connected to the bench **FRS 7.3.10**. We simulate session setup but not real keys.
- Certification. SignalSafe is a learning and pre-validation tool, not an approved facility.

Before every run: the pre-set

The FRS requires the equipment under test to be loaded and configured before a scenario starts **FRS 8.1**. SignalSafe's Bench Controller does this as an explicit, logged step so nothing is assumed:

SOFTWARE	STATION CONFIG	TRAIN CONFIG	LINKS	POWER-UP
Reference-model version pinned in the run record.	Yard, control table, track and gradient profile into REF-SVK.	Loco type, consist, brake parameters, Kavach parameters into REF-OVK.	RIUs and adjacent SVKs declared and known to the bench.	Units powered via the power adaptor, then the scenario clock starts.

Anatomy of a scenario

A scenario is the bench's basic unit **FRS 8.2.1**. Each SignalSafe scenario holds the four parts the FRS lists, plus two of our own:

- **Test sequence:** signalling inputs, timing, configuration.
- **Track description:** conditions and gradient profile.
- **Train description:** formation, loco type, rolling stock.
- **Simulation details:** which modules run, with what options.
- **Faults and expected observables** (SignalSafe additions).

The FRS puts **combining test cases into sequences** inside the bench's scope **FRS 6.1.1.6**. The TSE supports this directly:

- Scenarios can be **chained**: the end state of one (train position, mode, registration) becomes the start of the next.
- A **sequence** such as "enter territory → register → SPAD approach → radio loss → recovery" runs as one continuous simulation with one combined report.
- The live checks run across the whole sequence, not just each part.

Twelve scenarios, *each tied to a clause*

RDSO's official test cases (SIF 0533 v4.22) aren't in the documents we have. So the v1 library is our own: twelve scenarios written directly from SRS clauses, each with a setup, a trigger, and pass criteria the Evaluation Tool can check automatically.

S01 Stop short of a red

SIL-4 SRS 12 · 3.5.7.1

Loco approaches home signal S1 at ON. The loco pilot (DMI auto-player) keeps notching up.

PASS OVK brakes on the curve and stops short of S1: within 5 m in ≥90% of 200 seeded runs, within 30 m in ≥98%.

S02 Conflicting route inputs

SIL-4 SRS 12.1

The yard shows S4 OFF while the point for that route is not detected.

PASS SVK sends the most restrictive aspect for S4 and shortens the MA. No FS MA past S4.

S03 Head-on in block section

NON-SIL SRS 14.2 · 14.7

Two locos on TIN 201, approaching each other, both registered with SVK-A.

PASS SVK issues loco-specific SoS to both. Both brake immediately. Brakes release at 0 km/h **14.4**.

S04 Rear-end following

NON-SIL SRS 14.3

A faster loco closes on a slower one, same TIN, same direction.

PASS Only the rear loco brakes and stops ≥300 m behind. The front loco never brakes and its DMI never shows the alert.

S05 Radio goes silent

SIL-4 SRS 20.1 · 5.1

RMS drops all SVK→OVK packets from t = 40 s in an Absolute Block Section.

PASS At 6 s the aspect blanks but FS continues. At 30 s radio failure is declared and the pilot is asked to acknowledge. With no Ack within 15 s, FSB is applied.

S06 One radio dies

AVAILABILITY SRS 20.1.4

Radio 1 of the OVK fails in mid-section. Radio 2 stays healthy.

PASS The fault is logged and an NMS message is queued. Supervision continues normally, with no mode change and no brakes.

S07 Missing tags vs. a wrong tag

SIL-4 SRS 3.5.4.6 · 19.6

Run A: both tags of an expected pair are dropped. Run B: an unlinked tag is planted on the route.

PASS A: "tag missing" shown on the DMI after the L_doubt window, with no brakes. B: brakes on reading the unexpected tag.

S08 Roll back on a gradient

NON-SIL SRS 13.1

Loco stopped on a rising gradient with brakes released. It rolls opposite to the cab direction.

PASS Brake and audio-visual warning once roll-back exceeds 5 m. The intervention is shown on the DMI.

S09 Pulse-generator mismatch

SIL-4 SRS 20.5 · 6.5

PG-2 under-reads so the two PG speeds differ by more than 5 km/h for a full frame cycle.

PASS OVK goes to System Failure, logs the fault and queues an NMS alert.

S10 Signal flicker on approach

SIL-4 SRS 18.8 · 5.2

S4's ECR input chatters OFF→ON→OFF within 1 s while the train is 1.2 km out.

PASS SVK holds the MA for 2–5 s (configured value) and never sends an MA longer than the most restrictive state seen.

S11 LC gate auto-whistle

NON-SIL SRS 15.3–15.9

Approach to LC 42-A, in three variants: normal, MA ending short of the gate, and stopped at 400 m.

PASS Normal: continuous horn from 600 m to the gate. No horn when the MA is short of the gate or the train is at standstill. Ack alone cancels it.

S12 Odometry jump

NON-SIL SRS 20.7

The OVK's reported distance jumps 150 m between two position reports, inconsistent with its live speed.

PASS SVK detects the error beyond its 120 m default tolerance and sends a loco-specific SoS.

The SIL levels shown are those the SRS assigns to the function under test. They describe the real system, not SignalSafe. Thresholds marked "configurable" in the SRS are scenario parameters in SignalSafe, with the SRS default pre-filled.

What we can break, and *how we judge it*

A fault is a scheduled change to one bench module, triggered by time, location or another event. SignalSafe never alters the reference models' logic to cause a fault. It only changes what they see through their interfaces, just as a hardware bench would.

INTERFACE	INJECTABLE FAULTS	EXPECTED REACTION (SRS)
Radio (RMS)	Total loss, per-direction loss, % packet loss (seeded), added delay, radio hole over a stretch, one radio failed, replayed old frame	Aspect blank >6 s, failure at 30 s ABS / 10 s auto 20.1 . Old frame numbers discarded 17.12 .
RFID (RFID-S)	One tag of a pair missing, both missing, data mismatch in a pair, unexpected tag, one reader failed, both failed	Log or DMI only for misses. Brake on unexpected tag 19.6 . SF if both readers fail 20.2 .
Yard	Signal flicker, point not detected, track circuit stuck occupied, LSS OFF without line clear	MA held 2–5 s 18.8 . Most restrictive aspect 12.1 . Train-length abort after 3 min 8.1.6 .
Odometry (SS)	PG channel disagreement, location jump, gradual drift	SF at >5 km/h for a frame 20.5 . SVK SoS beyond 120 m 20.7 .
Brake (BIU-S)	Pressure feedback lost, traction cut-off feedback lost	OVK to System Failure 20.6 .
DMI · Time · Links	Active DMI link lost, GPS1/GPS2/PPS/RTC failures, TSRMS or adjacent-SVK link lost	SF 20.4 . GPS truth table 20.3 . SR authorization 20.8 .

How the Evaluation Tool decides

Following **FRS 7.6.15**, each scenario lists **expected observables**. An observable is an event type with filters plus a window in time, location, or both. After the run, the TET searches the combined log (TSL plus both units' event loggers) for each observable in order. A matched record is consumed, so the same brake event can't satisfy two steps. A step can depend on an earlier one, e.g. "within 15 s *after* the radio-failure prompt".

Results are **PASS** (observable missing or outside its window) or **FAIL** (a safety event that no step asked for, such as an EB nobody expected).

```
// S05 · expected observables (excerpt)
expect: [
  { id: 'blank', event: 'DMI_ASPECT_BLANK',
    train: 'L1', at: { after: 'fault:radio',
      sec: [6, 8] } },
  { id: 'fsStill', event: 'MODE', mode: 'FS',
    at: { after: 'blank', sec: [0, 1] } },
  { id: 'rfail', event: 'RADIO_FAILURE',
    at: { after: 'fault:radio', sec: [30, 32] } },
],
{ id: 'fsb', event: 'BRAKE_CMD', level:
  'FSB',
  at: { after: 'rfail', sec: [15, 17] } }
]
```

The causal chain

Every event the reference models emit carries a **causedBy** list naming the inputs or earlier events that triggered it. The report turns this into a readable chain. It is built directly from the log, not generated by AI, so it is exact and reproducible:

T 40.0 S · FAULT RMS drops SVK→OVK packets	T 46.0 S · 20.1.2 6 s silent → aspect blank, FS kept	T 70.0 S · 20.1.1 30 s with no packet → radio failure	T 70.0 S · 20.1.3 Degraded mode, Ack requested	T 85.0 S · 20.1.3 No Ack in 15 s → FSB applied
---	---	--	---	---

Four views, *one workbench*

Wireframes of the main screens, drawn before the build so layout decisions are made on paper first. The final UI follows the Signal Box identity: parchment panels, olive chrome and rust alerts.

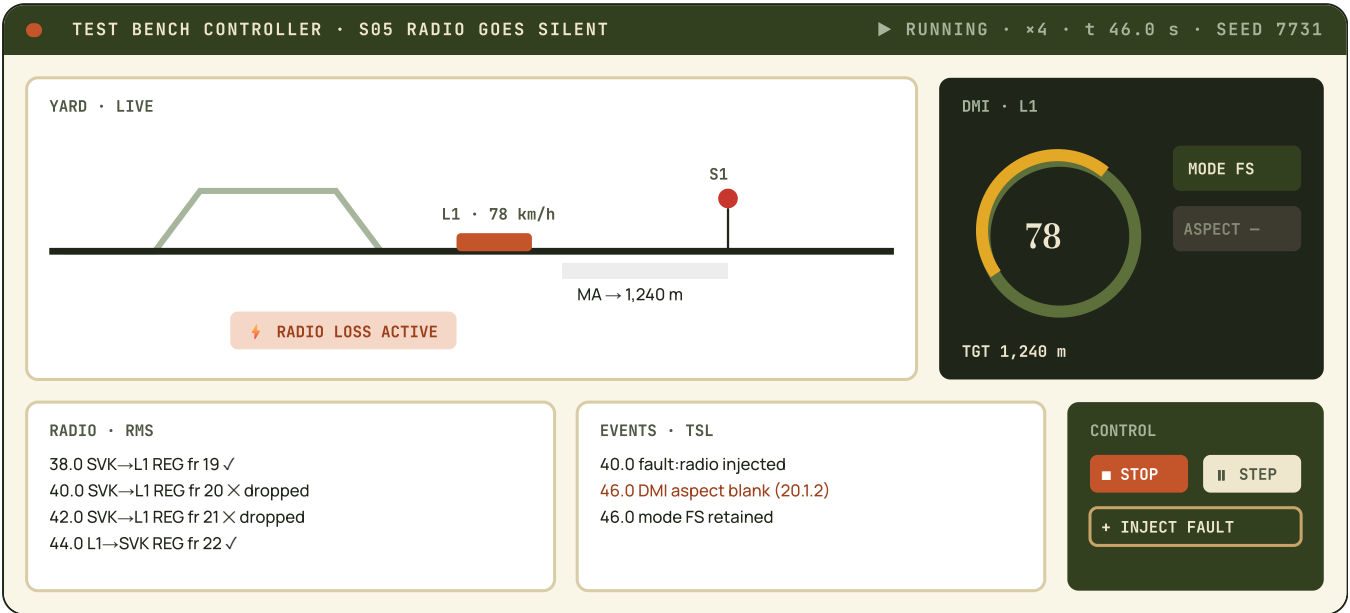


FIG. 2 · Bench Controller during a run: yard, mirrored DMI, radio and event streams, run controls
FRS 7.6.3.3–7.6.3.5.

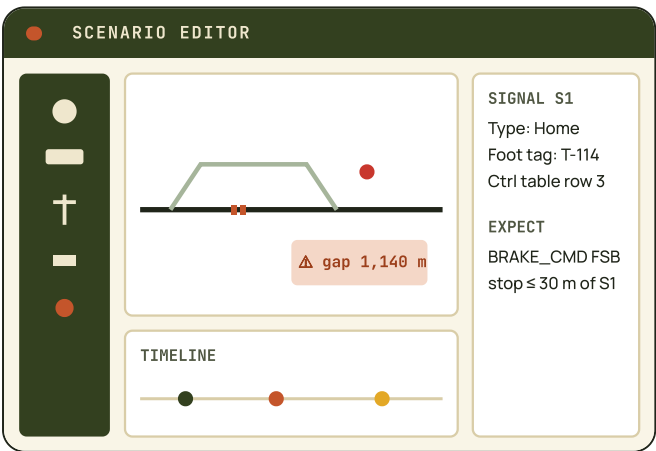


FIG. 3 · Scenario Editor: palette, yard canvas with live checks, timeline, inspector
FRS 7.6.2.

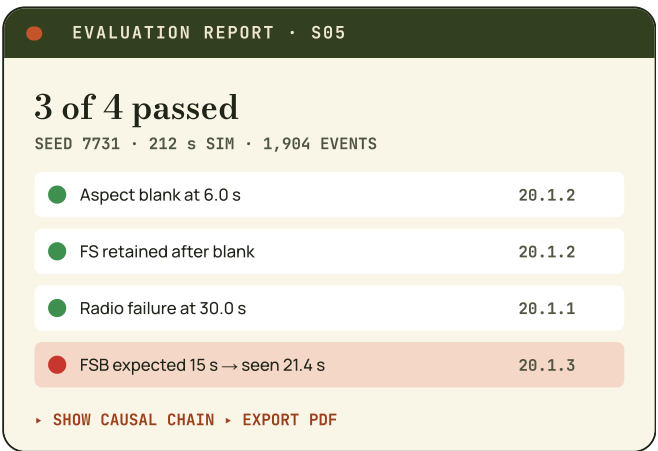


FIG. 4 · Evaluation report with clause per step (shown with a deliberately broken reference model)
FRS 7.6.15.

The DMI simulator, in detail

The DMI view follows the functions the SRS lists for the LP-OCIP **SRS 3.5.5.2**: current, permitted, target and section speed, overspeed, MA, target distance (to EOA, turnout, TSR/PSR, collision or SoS), signal aspect, mode, train length and configuration. Inputs: 12 soft keys, 4 navigation keys and the SoS, Common/Ack and Cancel buttons **3.5.5.3–3.5.5.4**. Health and SoS indicators are bi-colour green/red **3.5.5.8–3.5.5.10**. The exact screen layout is defined in Annexure B, which we don't have, so v1 uses our own layout with the same information and buttons.

Browser-first, *engine-centred*

The simulation engine is the product, and the UI is a window onto it. So the engine is plain TypeScript with no UI dependencies, runs in a Web Worker, and is tested on its own. Everything runs client-side, so v1 deploys as a static site.

LAYER	CHOICE	WHY
Language	TypeScript (strict)	Packet layouts, modes and events become types. Most protocol mistakes are caught at compile time.
Frontend	React 19 + Vite	Fast dev loop, and the same stack as our previous projects.
Yard rendering	SVG (editor) · Canvas 2D (live run)	SVG gives crisp, selectable elements for editing. Canvas keeps live runs smooth with many trains and fast-forward.
Engine	Pure TS in a Web Worker	Deterministic fixed-tick loop off the main thread. The UI receives state snapshots and log batches.
State	Zustand	Small and explicit. Engine snapshots replace whole slices, so there's no hidden mutation.
Validation	Zod schemas	Scenario files and adapter messages are validated at every boundary.
Storage	IndexedDB (Dexie) · JSON import/export	Scenario library and run logs stay in the browser. Scenarios are shareable as <code>.signalsafe.json</code> files.
Charts	Recharts	Speed and distance plots with the brake curve overlaid in reports.
Reports	Print-styled HTML → PDF	The same approach as this document, so reports match the house style.
Testing	Vitest · fast-check · Playwright	Unit tests per clause, property tests on safety invariants, end-to-end tests that run the twelve scenarios.
Hosting	Static (Vercel / GitHub Pages)	No server for v1. A backend arrives only with the shared library (section 11).

Why not a game engine or a backend simulation?

Unity or Godot would give nicer 3D, but SignalSafe is about logic and timing, not looks, and a browser tab is the lowest barrier for students. Running the engine server-side would make runs depend on network latency. That's acceptable for a dashboard but wrong for a tool that checks 2-second frame timing. A worker on the user's machine keeps every run exact.

Four layers, *straight from Figure 1*

The FRS reference architecture has a bench layer, a simulation layer, an adapter layer, and the equipment under test as a black box. SignalSafe keeps these four layers. The reference SVK and OVK models talk to the bench **only** through adapters, so they can later be swapped for real equipment.



Timing model

- **Fixed tick of 100 ms** simulated time. Dynamics, odometry and tag detection update every tick.
- **Radio frame of 2 s** = 20 ticks **SRS 17.11**. Each unit gets a slot in the frame. SVK and OVK packets carry frame numbers, and stale ones are discarded **17.12**.
- **Logging** every 2 s plus on events **SRS 21.2**, with time and location stamps.
- Runs go **step-by-step, real time, or up to ×20**. Simulated time is independent of wall-clock time, so results never depend on machine speed.

Tag detection at speed

At 200 km/h a train covers 5.6 m per 100 ms tick, more than the reader's ≤ 750 mm horizontal range **SRS 3.5.4.3**. So RFID-S doesn't sample position. Each tick it checks whether the antenna's path *crossed* a tag between the last position and this one. This gives no accidental misses at any speed, as **FRS 7.6.6.3** requires.

Core data model

```
Scenario { id, title, clauseRefs[], yard, trains[], timeline[], faults[], expect[], seed, params }
Yard      { tracks[], points[], signals[], tags[], lcGates[], tins[], towers[], controlTable[] }
Train     { locoId, config, brakeParams, lengthM, startTag, cabDirection, driver: 'auto' | 'manual' }
Fault     { kind, target, trigger: { atSec | atLocM | onEvent }, durationSec?, params }
LogEvent  { seq, tSim, frame, source, type, train?, locM?, data, causedBy: seq[] }
Observable { id, event, filters, at: { after?, sec?, locM? }, tolerance? }
```

Built the way a *safety tool* should be

1 • Deterministic, always

Same scenario + same seed = the same log, byte for byte. All randomness (packet loss, tag jitter) comes from one seeded generator, and nothing reads the wall clock. A failing run can always be replayed and debugged.

2 • Most restrictive wins

The reference models apply the SRS rule that, when several situations overlap, the most restrictive one governs **SRS 19.1–19.3**. When information stops, the last brake curve continues **19.4**. Unknown inputs default to the safe state.

3 • Invariants as tests

Property tests generate thousands of random yards and fault mixes and assert invariants on every tick. Examples: an FS MA never extends past a signal at ON. The front train in a rear-end case never brakes. A brake command is never weakened while its cause still holds.

4 • One clause, one test

Each SRS number SignalSafe enforces (6 s, 30 s, 15 s, 300 m, 5 m, 600 m, 5 km/h, 120 m...) lives in one config file with its clause reference and has at least one unit test. The appendix traceability table is generated from that file.

Honest limits

- **Not an approved test facility.** The FRS requires the real bench to be calibrated per Railway procedures **FRS 8.1.1.2**. SignalSafe has no such standing and its output is not evidence of compliance.
- **Reference models are our reading of the public SRS.** They are not any OEM's implementation and are not SIL-rated.
- **Several annexures aren't in our sources.** These are A1 (mode transitions), B (DMI layout), C (radio protocol and time slots), D (tag data format) and O (braking algorithm). In v1 those parts use simplified stand-ins, each labelled `STAND-IN` in the UI and in reports, so nobody mistakes them for the specification.
- **The braking model is simplified:** constant deceleration per brake level with a gradient term, until Annexure O is available.

Performance targets

TARGET	BUDGET	HOW WE'LL CHECK
Simulate a 10-minute, 4-train scenario	< 3 s at max speed	Benchmark in CI on the twelve scenarios
Live view frame rate at ×20	≥ 50 fps	Playwright trace on a mid-range laptop
Speed simulator vs. theoretical	≤ 1% deviation	Unit test against closed-form motion FRS 7.6.5.5
Bench-added radio latency	0 unless injected	Frame-cycle test stays within 2 s FRS 7.5.3.3

Eight weeks, *milestone by milestone*

Engine first, UI second, polish last. Each week ends with something runnable. Dates assume a start on 12 Oct 2026.

- 
- WEEK 0 • 08-11 OCT**
Documentation first. This specification, scenario list and clause table, signed off before code.
- WEEK 1 • 12-18 OCT**
Engine skeleton. Repo, TS project, fixed-tick loop, seeded RNG, event log with `causedBy`, yard graph model, Zod scenario schema.
- WEEK 2 • 19-25 OCT**
Train & track. Speed simulator with gradient and brake levels, swept-path RFID detection, linking with odometry correction, direction and TIN deduction **SRS 7, 16**.
- WEEK 3 • 26 OCT-01 NOV**
Reference SVK. Interlocking inputs → aspect and MA, control-table route check, most-restrictive rule, MA hold on flicker, registration and deregistration.
- WEEK 4 • 02-08 NOV**
Reference OVK + radio. Brake curve, FS supervision, SPAD, rollback, head-on and rear-end, RMS frames and slots, radio failure fallback. **Milestone: S01-S05 pass headless.**
- WEEK 5 • 09-15 NOV**
Workbench UI. Scenario Editor on SVG with live checks. Bench Controller with Canvas yard, run controls and streams.
- WEEK 6 • 16-22 NOV**
Operator panels. DMI simulator with auto-player, SMOCIP simulator, fault scheduler UI.
- WEEK 7 • 23-29 NOV**
Evaluation. TET matcher, causal chain view, speed-distance charts, PDF report. **Milestone: all twelve scenarios pass end-to-end.**
- WEEK 8 • 30 NOV-06 DEC**
Polish & ship. Accessibility pass, onboarding tour, sample library, performance budget, deploy, and the post-build version of this documentation with real screenshots.

Definition of done for v1

- All twelve scenarios pass, and each fails in the expected way when its reference-model rule is deliberately broken (mutation check).
- Every enforced number has a clause reference and a test. Every stand-in is labelled.
- A newcomer can build a two-station yard and run S01 in under ten minutes without help.

What could come *after vl*

Annexure-accurate models NEXT

Replace the stand-ins with A1, C, D and O once those annexures are available. Packet-level radio framing per Annexure C would let SignalSafe produce byte-accurate packets.

Hardware bridge RESEARCH

A small Node or Rust service that maps the adapter contract to RS-232/485 and Ethernet, which most bench interfaces use [FRS 7.6.1.17](#). The same scenarios could then drive real equipment. This needs OEM cooperation, as the FRS itself notes.

SIF 0533 import NEXT

If RDSO's official function test cases become available, import them into the TSE database as the FRS intends [FRS 7.6.2.2](#), alongside our own twelve.

Adjacent station handover NEXT

Full SVK-to-SVK soft handover per Annexure P, with up to six neighbours and shared border tags.

Shared scenario library LATER

A small backend (Node + Postgres) for publishing scenarios and reports, with reviewer comments on individual observables.

Classroom mode LATER

Instructors assign scenarios. Students predict the outcome before running ("will S05 brake, and when?"), then compare their prediction with the causal chain.

Plain-language explainer LATER

An optional assistant that reads the causal chain and the cited clauses and answers questions like "why didn't the front train brake?". It is limited to the run log and the SRS text, and always shows its sources.

NMS & TSRMS views LATER

A Network Monitoring System panel showing the queued NMS messages already modelled, and a TSRMS simulator for issuing temporary speed restrictions during a run.

Glossary, sources & *traceability*

Source documents

REF	DOCUMENT	NUMBER · STATUS
SRS	System Requirements Specification of KAVACH (The Indian Railway ATP), Version 4.0, Amendment 3. RDSO, S&T Directorate	RDSO/SPN/196/2020 · eff. 07.06.2024
FRS	Draft Functional Requirement Specification for KAVACH Generic Test Facility, Version 1.0. RDSO, S&T Directorate	RDSO/GTB Kavach/FRS/2026 · draft, Jul 2026
SIF	Function Test Cases at System Level for KAVACH, Part I & II (referenced by the FRS, not available to us)	SIF 0533 v4.22

Enforced values (traceability excerpt)

RULE	DEFAULT	CLAUSE	SCENARIO
Radio frame cycle / location report interval	2 s	17.11 · 9.2	all
Aspect blanked when last SVK packet older than	6 s	20.1.2	S05
Radio failure declared (Absolute / Automatic block)	30 s / 10 s	20.1.1	S05
Loco pilot acknowledgement window before brake	15 s	20.1.3	S05
SVK holds FS MA after OVK silence / aspect change / flicker	6 s · 2 s · 2-5 s	5.1 · 5.2 · 18.8	S10
SVK deregisters silent loco (Absolute / Automatic)	2 min / 30 s	3.4.5.3	S05
Rear-end stopping margin behind train ahead	300 m	14.3 · 3.5.6.4	S04
Roll-back before intervention	5 m	13.1	S08
Stop accuracy at EOA (90% / 98% of cases)	≤5 m / ≤30 m	3.5.7.1	S01
PG speed disagreement for one frame → SF	>5 km/h	20.5.1	S09
Odometry error tolerance before SVK SoS	120 m	20.7.1	S12
LC auto-whistle start distance	600 m	15.8	S11
Max gap between normal tags · odometry error bound	1000 m · 5 m + 5%	3.4.2.6 · 3.4.2.4	TSE check
Radio-hole distance computed by OVK	hole + 10 s × v	3.4.8.7(g)	v2
Train length accuracy / update threshold	±25 m / 25 m	8.1.3-8.1.4	v2

Glossary & *design language*

ATP	Automatic Train Protection	MA · EOA	Movement Authority · End of Authority
SVK · OVK	Stationary · Onboard Kavach	TIN	Track Identification Number
LP-OCIP · DMI	Loco Pilot Operation cum Indication Panel · Driver Machine Interface	SMOCIP	Station Master Operation cum Indication Panel
BIU	Brake Interface Unit	NB · FSB · EB	Normal · Full-Service · Emergency brake
RIU	Remote Interface Unit	PG	Pulse Generator (speed sensor)
NMS	Network Monitoring System	TSRMS	Temporary Speed Restriction Management System
PSR · TSR	Permanent · Temporary Speed Restriction	LSS	Last Stop Signal
SoS	Emergency stop message (manual or generated)	SIL	Safety Integrity Level (CENELEC)
ABS	Absolute Block Section	L_doubt	Odometry uncertainty window for tag expectation

Design language: "Signal Box"

One visual identity throughout, drawn from vintage railway print: **olive and sage greens** on **parchment**, **rust** for anything that needs attention, and brass hairlines with ring-and-dot ornaments. Torn-paper edges separate scenes. Type is **Bodoni Moda** for display, **Manrope** for reading and **JetBrains Mono** for clause numbers, packet fields and timings. Signal colours (●●●) are reserved for real signal aspects and pass/fail states, never for decoration.